1. (a) Briefly discuss about the role of driver while configuring a layered testbench to verify any DUT. Clearly discuss about input and output flow for “Driver” component.

   (b) Which features of the System Verilog make it preferable over Verilog HDL when verification of a DUT is the main objective? Briefly discuss about them?

2. Write a SV program to create a 2D array with the following rows: {14, 10, 15, 3}, {12, 30, 16, 9}, and {7, 9, 15, 10}. Extract elements from this 2D array that are divisible by 3 but not divisible by 2 and store them in a dynamic array. Finally, display the contents of the original 2D array and the resulting dynamic array, using loops and conditional statements for processing.

3. Write a SV program to process the string str1 = "We are appearing for CAT examination". Create two new strings: str2, containing all the vowels, and str3, containing all the consonants from str1, ignoring spaces. Compare the sizes of str2 and str3, and display which string has more characters and the difference in size. Additionally, count and display the total number of spaces in str1.

4. Write a SV program to assign numbers {3, 6, 4, 5, 2} to five fruits {Apple, Orange, Guava, Grapes, and Mango} using an enumerated data type. Use typedef to give the enumerated data type a new name, fruits. Create a variable of this type and assign Grapes as its current value. Display the values of the first and last members of the enumeration, and also display the name of the current value (i.e., Grapes).

5. **(a)** Write a SV program to create a queue (q1) of string elements {“AB”, “BC”, “CA”, “CB”, “BA”}. Declare two additional queues (q2 and q3) and store the first two and last three elements of q1 into q2 and q3, respectively. Add two more elements {“CD”, “DC”} to q2 and {“EF”, “FE”} to q3. Finally, display the elements of all three queues, excluding the first element of each queue.

   (b) With appropriate examples, discuss the difference between packed and unpacked arrays. How size of a dynamic array is increased during run time?
6. Write a program to create a dynamic array containing the first 15 multiples of 7 as its elements. Using the "with" clause and built-in functions, perform the following tasks (use the elements of the created array):
   (a) Store the elements, which are greater than 20 but less than 80, into a queue and display it.
   (b) Store and display the indices of the elements that are divisible by 5.
   (c) Store the indices and values of all odd numbers into two separate queues and display their items.
7. There are two strings given as str1 = “We are students of VIT university” and str2 = “This is HDL verification”. Write a program to count number of vowels in both the strings. Then, using conditional statement, check whether str1 or str2 has more number of vowels. Display the difference between their vowel counts as well.
8. Write a SV program to declare an enum named “Colors” with the following members and values: red=0, green, blue=4, yellow, white=10, and black. Use a for loop with the first and next methods to display the name and value of each color. Additionally, implement functionality to check if a specific color (e.g., yellow) exists in the “Colors” enum and display a message indicating whether it exists or not.

9. Design a SV program that keeps track of quiz scores for students. Use an associative array where each key is a student's name (string) and the value is their quiz score (int). Traverse the array using a loop to determine and display the names of the students with the highest and lowest scores.
10. Write a SV program using qa = {3, 6, 9, 12} and qb = {2, 4, 6, 8}. Create qc using the first 3 elements of qa and last of qb. Insert 50 at the beginning of qc and delete its 3rd element. Create qd by merging qa and qb in reverse, alternating elements. Display all queues after each step.

Guidelines for the preparation of the question paper

The question paper setter must keep in mind the following instructions before preparing the question paper:

1. BL: Bloom's Taxonomy Levels (1: Remembering, 2: Understanding, 3: Applying, 4: Analysing, 5: Evaluating, 6: Creating)
   Note: BL 1 and BL 2 = 40% of the questions; BL 3 = 30% of the questions; and BL 4 = 30% of the questions.
   BL: Coursewise deviations in percentage of BLs are permitted. As per academic regulations, upto 80% of HOT questions are permitted.

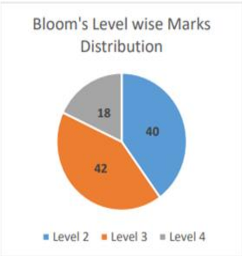
2. Fill out the appropriate CO (course outcomes) for each question, as it is vital for CO attainment calculation. Refer to the syllabus for course outcomes.

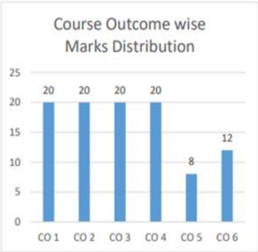
3. Difficulty Level: Easiness of the question in the grade Easy, Average, Tough, or E/A/T. The recommended proportion of the questions should be 25%, 50%, and 25%, respectively, to ensure a normal distribution.

4. CO & BL Mapping: Prepare a pie chart and bar diagram for the BL vs. Marks and CO vs. Marks.

For Example:

Qs No	BL	CO
1		
2		
3		
4		
5		





BL – Bloom's Taxonomy Levels (1- Remembering, 2- Understanding, 3 – Applying, 4 – Analysing, 5 – Evaluating, 6 - Creating)
CO – Course Outcomes

   Faculty can work out this bar diagram and pie chart and check the balance of questions in the given format before uploading the question paper.

5. Ensure that you have followed the approved syllabus.
6. Set the question paper for the full syllabus, covering all the units uniformly.
7. Use the given template for the preparation of the question paper.
8. Under each question, sub-sections can be included.
9. General Instructions indicate the instructions for the whole of the question paper.
10. Give the question number continuously, starting with '1', and do not break continuity when going to the next section.
11. Module No.: Type the module number as given in the syllabus, like 1, 2, 3, 4, 5.
12. A detailed scheme of evaluation is mandatory along with the question paper. Without the key, the question paper will not be accepted. The same will be made accessible to students immediately after the completion of the examination for reference and possible re-evaluation.
13. If your relative is appearing for this examination, please decline the offer and return the documents.

14. All paper setters are requested to keep their appointments strictly confidential. including your passwords
15. The question paper should be such that a candidate of decided ability and good preparation in that course can reasonably be expected to answer completely within the allotted time.
16. Ensure that the questions are not used in previous examinations or anywhere else, like a model question paper.
17. Ensure that the questions are balanced in terms of syllabus coverage, marks, and time taken to answer each question.
18. Follow the time schedule since many follow-up activities like moderation, formatting, and printing out the question paper rely on your punctuality.
19. The undertaking should be submitted along with question paper.

UNDERTAKING

1. I hereby certify that the question paper was typed by me and I have not retained any copy of the same in any form.

2. I hereby certify that I have taken sufficient care to destroy the draft copy used and have deleted the relevant files from my computer taking care to ensure it is not retrievable by any means.

3. I hereby certify that I have taken utmost care to maintain confidentiality.

4. I am certifying that none of my relatives are either studying for or appearing for the examination for which the question paper has been set by me.

5. Any temporary storage is completely protected with passwords. Also, no hard copy was retained with me.

	Question paper setter	Moderator 1	Moderator 2
Name:	Dr. Chandan Kumar Pandey	Dr. Ameet Chavan	____________________
Emp ID:	70155	__________________	___________________
Signature with date		__________________	___________________